



Agriculture: Apple Pest Management Guidelines

Apple Scab

Venturia inaequalis

Symptoms and Signs

Superficial, velvety dark-olive-to-black [spots\(/PMG/V/D-AP-VINA-FO.007.html\)](#) on fruits and leaves are the initial symptoms of apple scab. These fruit spots become scablike with age and tissues may become misshapen.

Comments on the Disease

The fungus survives in dead leaves on the ground. [Primary spores\(/PMG/FIG/app-f59.html\)](#) are discharged during spring rains and [infect young leaves and fruits\(/PMG/V/D-AP-VINA-BL.004.html\)](#) during prolonged moisture. Secondary spores may spread disease from established infections on trees. A lack of spring rains in most years reduces its importance in California in the southern Central Valley, but as a rule, this disease requires yearly treatments in coastal and foothill districts. Ornamental crab apples are also hosts. As plant parts mature and the weather gets warmer, susceptibility to this disease decreases, but pinpoint scab can occur during extended periods of moisture during summer.

Management

The main objective in scab management is the reduction or prevention of primary infections in spring. Extensive primary infections result in poor fruit set and make scab control during the season more difficult. If primary infections are successfully controlled, secondary infections will not be serious. The key to success in scab control is exact timing and full coverage. Wet periods, temperature, and relative humidity are important factors. Because scab control often is part of a combination treatment aimed at other diseases and insect control, choice of materials and timing are also extremely important.

A fall foliar fertilizer application of zinc sulfate and urea hastens leaf fall and speeds decomposition of fallen leaves. This in turn reduces the level of overwintering apple scab inoculum.

Organically Acceptable Methods

Treatments with lime and sulfur, sulfur alone, or with approved fixed copper products are organically acceptable.

Monitoring

Apple scab outbreaks can be predicted based on temperature and moisture conditions. The table below, derived from research by Mills and La Plante, gives hours needed at various temperatures under constantly wet conditions for primary spores (ascospores) to cause infection in spring. This system for forecasting scab and timing sprays has been validated for apple-growing regions in the northern areas of California, but not for the central and southern San Joaquin Valley where temperatures tend to be higher in spring than those in the table.

TABLE 1. Mills and La Plante Table.

HOURS OF WETTING REQUIRED FOR INFECTION	
Average temperature (degrees °F)	From primary inoculum (ascospores) (hours)
78	13
77	11
76	9.5
61–75	9
60	9.5
57–59	10
55–56	11
54	11.5
52–53	12
51	13
50	14
49	14.5
48	15
47	17
46	19
45	20
44	22
43	25
42	30
33–41	*

* Not known

How to use the table: Figure the average temperature for the rain period by adding the maximum and the minimum temperatures and dividing by 2. If wet periods are intermittent, total their durations until there is a period of at least 6 hours of continuous dryness. You will need a wetness recorder to do this efficiently. If the dry period is sunny, and drying is quick and thorough, it is assumed that 6 hours after the trees have dried, the danger is passed. If drying is slow, and humidity remains high, then the 6-hour dry period is extended by a safety margin of 3 to 4 hours.

Treatment Decisions

Unless wetness periods are being monitored as outlined in the section above, apply protective fungicides at regular intervals beginning with [green tip\(/PMG/R/S-AP-REPG-B3.007.html\)](#). Be sure to monitor wetness periods throughout the spring to ensure that trees are adequately protected.

PERIOD OF EFFECTIVENESS ¹		
FUNGICIDE	When used as protectant (days)	When used as kickback ² (hours)
Captan	10	0
Copper	7–10	—
Dithane	10	0
Flint	7–10	100
Lime sulfur	5–7	36
Procure	7–10	72
Rally	14	96
Rubigan	14	96
Scala	7–10	—
Sovran	7–10	96
Sulfur (micronized)	5–7	0
Topsin M	10	36–48
Vangard	7–10	48
Ziram	7–10	0

—	Information not available.
¹	Where range of days or hours is given, the difference reflects application rates –higher rates offer longer protection.
²	Eradicant fungicides have systemic action. Some are translocated within the host tissue and are able to kill the scab fungus up to a certain length of time after infection occurs. This is called the kickback period. Because kickback periods may change, always check the label for the most recent information. Kickback is calculated from the <i>beginning</i> of an infection period, as determined by the Mills and La Plante table.

Common name	Amount to use**		REI‡	PHI‡
(Example trade name)	(conc.)	(dilute)	(hours)	(days)
Pesticide precautions Protect water Calculate VOCs Protect bees				
Not all registered pesticides are listed. The following are ranked with the pesticides having the greatest IPM value listed first —the most effective(/agriculture/apple/Fungicide-Efficacy-for-Apple-Diseases/) and least likely to cause resistance are at the top of the table. When choosing a pesticide, consider information relating to the pesticide's properties(/agriculture/apple/General-Properties-of-Fungicides-Used-in-Apples/) and application timing(/agriculture/apple/Treatment-Timings-for-Key-Diseases/) , honey bees(/bee-precaution-pesticide-ratings/) , and environmental impact(/mitigation/) . Always read the label of the product being used.				

Common name		Amount to use**		REI‡	PHI‡
(Example trade name)		(conc.)	(dilute)	(hours)	(days)
<i>DORMANT</i>					
A.	LIQUID LIME SULFUR#	5–10 gal/100 gal		48	0
	... <i>PLUS</i> ...				
	WETTABLE SULFUR#	4–6 lb/100 gal		24	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M2)				
	COMMENTS: Use wettable sulfur alone after full bloom(/PMG/R/S-AP-REPG-B9.005.html) as lime sulfur may burn foliage if applied during warm weather. Lime sulfur is incompatible with most other pesticides. Check before use. Do not apply within 3 weeks of an oil application.				
FOLIAGE SPRAY					
A.	FENARIMOL				
	(Vintage SC)	9–12 oz/acre		24	30
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Demethylation inhibitor (3)				
	COMMENTS: Do not apply between popcorn and full bloom. Do not apply more than 84 fl oz/acre per season.				
B.	MYCLOBUTANIL				
	(Rally 40WSP)	Label rates		24	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Demethylation inhibitor (3)				
	COMMENTS: Apply 400 gal/acre. Do not apply between popcorn and full bloom. During periods favorable for scab development, use high rate at a 7 day application interval. For postbloom control, can be used in combination with another protectant fungicide. Do not apply more than 5 lb/acre per season or graze or feed cover crops grown in treated areas to livestock. For application by ground equipment only.				
C.	TRIFLUMIZOLE				
	(Procure 480 SC)	8–16 fl oz/acre		12	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Demethylation inhibitor (3)				
	COMMENTS: Do not exceed 16 oz/acre per application.				
D.	TEBUCONAZOLE/TRIFLOXYSTROBIN				
	(Adament 50WG)	4–5 oz/acre	12		75
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Demethylation (3) and Quinone outside inhibitor (11)				
	COMMENTS: Most effective when applied before a rainfall and allowed to dry. Apply at 5–10% bloom and make a second application at 80–100% bloom.				

Common name		Amount to use**		REI‡	PHI‡
(Example trade name)		(conc.)	(dilute)	(hours)	(days)
E.	TRIFLOXYSTROBIN				
	(Flint)	2–2.5 oz/acre		12	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Quinone outside inhibitor (11)				
	COMMENTS: Alternate with a sterol inhibitor fungicide. Do not apply more than 2 consecutive applications before alternating. Do not apply more than 11 fl oz/acre per season.				
F.	DIFENOCONAZOLE/CYPRODINIL				
	(Inspire Super)	12 fl oz/acre		12	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Demethylation (3) and Anilinopyrimidine (9)				
	COMMENTS: Most effective when applied before a rainfall and allowed to dry. Apply at 5–10% bloom and make a second application at 80–100% bloom.				
G.	FLUOPYRAM/TRIFLOXYSTROBIN				
	(Luna Sensation)	4.0–5.8 fl oz/acre		12	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Succinate dehydrogenase inhibitors (7) and Quinone outside inhibitor (11)				
	COMMENTS: Most effective when applied before a rainfall and allowed to dry. Apply at 5–10% bloom and make a second application at 80–100% bloom.				
H.	PYRACLOSTROBIN/BOSCALID				
	(Pristine)	14.5–18.5 oz/acre		12	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Quinone outside inhibitor (11) and Carboxamide (7)				
	COMMENTS: A strobilurin/carboxyanilide fungicide.				
I.	PYRIMETHANIL				
	(Scala SC)	7–10 fl oz/acre		12	72
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Anilinopyrimidine (9)				
	COMMENTS: Make application on 7 day or longer, depending on disease conditions. Do not apply more than 40 fl oz product/acre per crop.				
J.	KRESOXIM-METHYL				
	(Sovran)	3.2–6.4 oz/acre		12	30
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Quinone outside inhibitor (11)				

Common name		Amount to use**		REI‡	PHI‡
(Example trade name)		(conc.)	(dilute)	(hours)	(days)
	COMMENTS: Begin applications at 1/2 inch green or when conditions are conducive to disease development. Repeat at 7- to 10-day intervals. Do not apply more than 25.6 oz/acre per season.				
K.	THIOPHANATE-METHYL				
	(Topsin-M 70WP)	1–1.5 lb/acre		48	1
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Methyl benzimidazole carbamates (1)				
	COMMENTS: Apply at 5- to 10-day intervals from green tip through petal fall; continue at 7-to 14-day intervals as needed. Do not apply more than 4 lb product/acre per season.				
L.	CYPRODINIL				
	(Vangard WG)	3–5 oz/acre		12	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Anilinopyrimidine (9)				
	COMMENTS: Make applications on a 7- to 10-day interval. At the 3 oz (protectant) rate there is no kickback period of control, whereas at the 5 oz rate this material has a 48 hour kickback period.				
M.	CAPTAN				
	(Captan 50WP)	4–8 lb/acre		24	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M4)				
	COMMENTS: Apply 400 gal/acre. Do not apply in combination with, immediately before, or closely following oil sprays.				
N.	LIQUID LIME SULFUR#	2–3 gal/100 gal		48	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M2)				
	COMMENTS: Lime sulfur is incompatible with most other pesticides. Check before use. May burn foliage if applied during warm weather. Do not apply within 3 weeks of an oil application.				
O.	MICRONIZED SULFUR#				
	(Thiolux, Microthiol, etc.)	Label rates		24	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M2)				
	COMMENTS: Can be applied after bloom. Do not apply within 3 weeks of an oil application.				
P.	MANCOZEB				
	(Dithane DF)	3 lb/acre		24	77
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M3)				

Common name		Amount to use**		REI‡	PHI‡
(Example trade name)		(conc.)	(dilute)	(hours)	(days)
	COMMENTS: Begin applications at 0.25- to 0.5- inch green tip and continue at 7- to 10-day intervals. Do not exceed 21 lb/acre per year of Dithane.				
Q.	ZIRAM				
	(Ziram 76DF)	6 lb/acre		48	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M3)				
	COMMENTS: Apply from pre-bloom through cover sprays as needed. Do not apply more than 32 lb/acre per year.				
R.	FIXED COPPER#	Label rates		24	0
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M1)				
	COMMENTS: Apply at bud break using high label rate; later applications should be at low label rates. May cause russetting at rates over 0.5 lb/acre. Not all copper compounds are approved for use in organic production; be sure to check individual products.				

‡	Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without protective clothing. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases the REI exceeds the PHI. The longer of two intervals is the minimum time that must elapse before harvest.
#	Acceptable for use on organically grown produce.
¹	Group numbers are assigned by the Fungicide Resistance Action Committee (FRAC)(http://www.frac.info/) according to different modes of actions. Fungicides with a different group number are suitable to alternate in a resistance management program. In California, make no more than one application of fungicides with mode-of-action Group numbers 1,4,9,11, or 17 before rotating to a fungicide with a different mode-of-action Group number; for fungicides with other Group numbers, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action Group number.



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